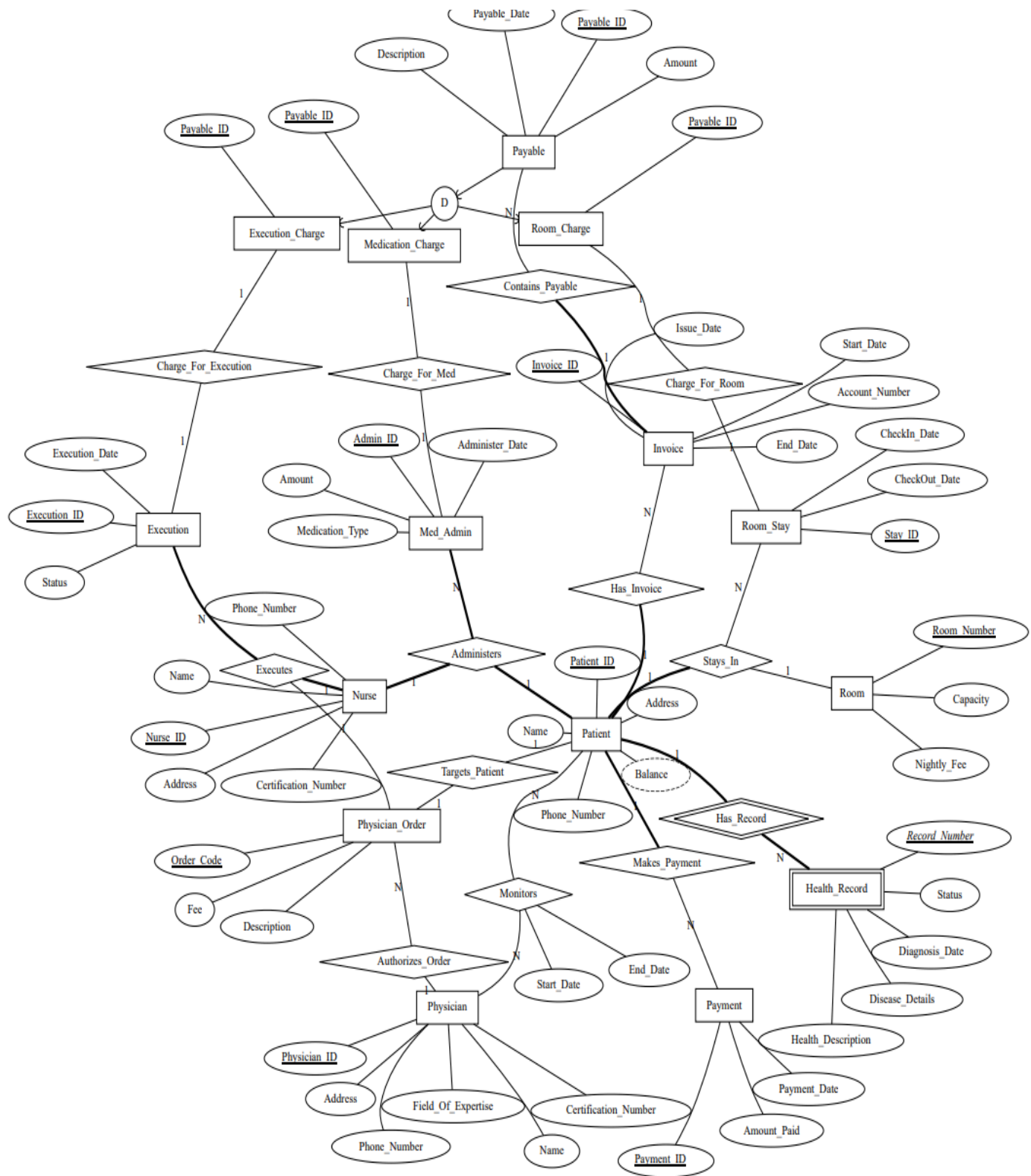


Final Report

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Assumptions:

- For Part 1 there were lots of assumptions that needed to be made. For instance, in a hospital it is common for a patient to come in multiple times so we needed to account for a patient having multiple health records. Then, room numbers are never reused in a hospital they are unique and only appear once so we needed to assume this. Then in Patient we know that balanced is a derived attribute so it has to be calculated so we had to assume that. Then moving to relationships in the ERD, we assume that a physician can care for multiple patients but a patient can be cared by multiple physicians. Then we had to assume that a patient would bring guests so we had to make sure we had something for that. Then I assumed since in most hospitals nurses are the ones who administer medicine so I assumed it would be the same here. Other than that I think I was able to come up with the rest based on the information provided to us.
- For Part 2, not much was needed to assume here. Since we know what our ERD notation is so coming up with the set of relations was easy. Then for the tables it was easy too you just needed to make sure you made it so that it can handle any information you inserted. Then for the inserts I made my own information based on the tables and made sure I made enough inserts for each table.
- For Part 3 I was a bit stuck on creating queries. Since we are told to insert information into the tables before the queries, I wasn't able to make sure all my queries were about the information I made so that is why some of them are null. So I assumed that if someone were to use this and insert more stuff then these queries would be useful to the database. I assumed that we could use SIGNAL SQLSTATE since I wanted to have error messages in my queries. I did ask TA's and they said others asked and it was fine but I had to make assumptions. Then in the queries the only thing I assumed were the text inserted like medication amounts or status of patient since I had to assume it would use the inserts I did.



Queries:

JOIN Queries -

```
4      -- Query 1: This JOIN query gets the amount of medication given to a patient and it shows when and who
5  •    SELECT n.Name AS Nurse_Name, pat.Name AS Patient_Name, m.Medication_Type, m.Amount, m.Administer_Date
6      FROM Medication_Administration m
7      JOIN Nurse n ON m.Nurse_ID = n.Nurse_ID
8      JOIN Patient pat ON m.Patient_ID = pat.Patient_ID;
9
```

Result Grid Filter Rows: Export: Wrap Cell Content:					
	Nurse_Name	Patient_Name	Medication_Type	Amount	Administer_Date
▶	Ava	PatOne	Ibuprofen	100mg	2025-01-12
	Olivia	PatTwo	Antibiotic	250mg	2025-02-03
	Ben	PatThree	Insulin	4 units	2025-03-14
	Oscar	PatFour	ChemoDrug	30mg	2025-04-07
	Riazi	PatFive	Vaccine	3 dose	2025-05-02

```
12     -- Query 2: This JOIN Query lists which physicians are monitoring patients, but also sh
13  •    SELECT phys.Name AS Physician_Name, pat.Name AS Patient_Name, m.Start_Date, m.End_Date
14     FROM Monitors m
15     JOIN Physician phys ON m.Physician_ID = phys.Physician_ID
16     JOIN Patient pat ON m.Patient_ID = pat.Patient_ID;
```

Result Grid Filter Rows: Export: Wrap Cell Content:				
	Physician_Name	Patient_Name	Start_Date	End_Date
▶	Dr. A	PatOne	2025-01-10	2025-01-15
	Dr. B	PatTwo	2025-02-01	2025-02-05
	Dr. C	PatThree	2025-03-12	2025-03-18
	Dr. D	PatFour	2025-04-05	2025-04-12
	Dr. E	PatFive	2025-05-01	2025-05-03

```
16     -- Query 3: This JOIN query lists the amount the patient has paid and showcases
17  •    SELECT pay.Payment_Date, pay.Amount_Paid, pat.Name AS Patient_Name, i.Issue_Date
18     FROM Payment pay
19     JOIN Patient pat ON pay.Patient_ID = pat.Patient_ID
20     LEFT JOIN Invoice i ON pat.Patient_ID = i.Patient_ID;
```

Result Grid Filter Rows: Export: Wrap Cell Content:				
	Payment_Date	Amount_Paid	Patient_Name	Issue_Date
▶	2025-01-25	340.00	PatOne	2025-01-20
	2025-02-15	420.00	PatTwo	2025-02-10
	2025-03-25	510.00	PatThree	2025-03-20
	2025-04-20	660.00	PatFour	2025-04-15
	2025-05-10	210.00	PatFive	2025-05-05

AGGREGATION Queries -

```

23      -- Query 1:
24      • SELECT Medication_Type, COUNT(*) AS Usage_Count
25      FROM Medication_Administration
26      GROUP BY Medication_Type;
27

```

Medication_Type	Usage_Count
Ibuprofen	1
Antibiotic	1
Insulin	1
ChemoDrug	1
Vaccine	1

```

28      -- Query 2: This aggregate query gets the total amount per patient. This is helpful by showing how much each patient has to pay overall.
29      • SELECT i.Patient_ID, pat.Name AS Patient_Name, SUM(pay.Amount) AS Total_Invoice_Amount
30      FROM Invoice i
31      JOIN Payable pay ON i.Invoice_ID = pay.Invoice_ID
32      JOIN Patient pat ON i.Patient_ID = pat.Patient_ID
33      GROUP BY i.Patient_ID, pat.Name;

```

Patient_ID	Patient_Name	Total_Invoice_Amount
1	PatOne	860.00
2	PatTwo	360.00
3	PatThree	550.00
4	PatFour	880.00
5	PatFive	900.00

```

35      -- Query 3:
36      • SELECT Description, COUNT(Order_Code) AS Total_Orders, AVG(Fee) AS Average_Fee
37      FROM Physician_Order
38      GROUP BY Description;

```

Description	Total_Orders	Average_Fee
Blood	1	50.000000
X-Ray	1	100.000000
MRI Scan	1	200.000000
Chemo	1	400.000000
Vaccination	1	10.000000

NESTED Queries -

```

40  -- NESTED Queries
41  -- Query 1: This nested query gets
42  • SELECT Nurse_ID, Name
43      FROM Nurse
44  WHERE Nurse_ID IN (
45      SELECT Nurse_ID
46      FROM Medication_Administration
47      GROUP BY Nurse_ID
48      HAVING COUNT(*) = 1
49  );

```

Result Grid		Filter Rows:	Edit
Nurse_ID	Name		
66	Ava		
77	Olivia		
88	Ben		
99	Oscar		
100	Riazi		

```

51  -- Query 2: This query returns the most expensive order that a patient g
52  • SELECT phys.Order_Code, phys.Description, phys.Fee, phys.Patient_ID
53      FROM Physician_Order phys
54  WHERE phys.Fee > ALL (
55      SELECT Fee
56      FROM Physician_Order
57      WHERE Patient_ID = phys.Patient_ID AND Order_Code <> phys.Order_Code
58  );
59

```

Result Grid					Filter Rows:	Edit	Export/Import:	Wrap C
Order_Code	Description	Fee	Patient_ID					
100	Blood	50.00	1					
111	X-Ray	100.00	2					
222	MRI Scan	200.00	3					
333	Chemo	400.00	4					
444	Vaccination	10.00	5					

```

61 • SELECT Name
62 FROM Nurse
63 WHERE Nurse_ID NOT IN (
64     SELECT DISTINCT Nurse_ID
65     FROM Medication_Administration
66 );
67

```

Result Grid	Filter Rows:	Exp
Name		

OTHER Queries -

```

66 -- Query 1:
67 • SELECT n.Name AS Nurse_Name, e.Status, phys.Description
68 FROM Execution e
69 JOIN Nurse n ON e.Nurse_ID = n.Nurse_ID
70 JOIN Physician_Order phys ON e.Order_Code = phys.Order_Code
71 WHERE e.Status = 'Pending';
--

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content: IA
Nurse_Name	Status	Description	
Ava	Pending	Blood	
Oscar	Pending	Chemo	

```

73 -- Query 2: This query shows how long a patient stays in a room. This is helpful because we are
74 • SELECT rms.Patient_ID, pat.Name, DATEDIFF(rms.CheckOut_Date, rms.CheckIn_Date) AS Stay_Length
75 FROM Room_Stay rms
76 JOIN Patient pat ON rms.Patient_ID = pat.Patient_ID;
77

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content: IA
Patient_ID	Name	Stay_Length	
1	PatOne	5	
2	PatTwo	4	
3	PatThree	6	
4	PatFour	7	
5	PatFive	2	

```

78 -- Query 3: This query gets the total amount that physicians charged. Thi
79 • SELECT phys.Name AS Physician_Name, SUM(physo.Fee) AS Total_Fees_Charged
80 FROM Physician phys
81 JOIN Physician_Order physo ON phys.Physician_ID = physo.Physician_ID
82 GROUP BY phys.Name;
--

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

Physician_Name	Total_Fees_Charged
Dr. A	50.00
Dr. B	100.00
Dr. C	200.00
Dr. D	400.00
Dr. E	10.00

```

84 -- Query 4: This query finds patients who stay in expensive rooms. This
85 • SELECT DISTINCT pat.Name AS Patient_Name, rm.Room_Number, rm.Nightly_Fee
86 FROM Room_Stay rms
87 JOIN Room rm ON rms.Room_Number = rm.Room_Number
88 JOIN Patient pat ON rms.Patient_ID = pat.Patient_ID
89 WHERE rm.Nightly_Fee = (
90     SELECT MAX(Nightly_Fee) FROM Room
91 );
92

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

Patient_Name	Room_Number	Nightly_Fee
PatFive	5	800.00

```

94 • SELECT Name, Patient_ID
95 FROM Patient
96 WHERE Patient_ID IN (
97     SELECT Patient_ID
98     FROM Health_Record
99     GROUP BY Patient_ID
00     HAVING COUNT(DISTINCT Disease_Details) > 1
01 );

```

Result Grid |   Filter Rows: | Edit:   

Name	Patient_ID
PatTwo	2

```
104 • SELECT pat.Patient_ID, pat.Name
105 FROM Patient pat
106 WHERE pat.Patient_ID NOT IN (
107     SELECT DISTINCT Patient_ID FROM Physician_Order
108 );
```

Result Grid			Filter Rows:	Edit: 			Export
	Patient_ID	Name					
+	NULL	NULL					